
3AL Exam Syllabus

The exam will cover all of the content covered in lectures. Chapter 1 was just review, so you only need to study Chapters 2 to 5. The best practice for the exam is doing tutorials (not past papers). The tutorials contain several questions on each topic, a past paper contains maybe one or two on each topic. Note that none of the past exams cover rings, but rings will be included in this year's exam.

Here's a brief outline of the topics we've covered:

1. Isomorphism Theorems
 - (a) First Isomorphism Theorem
 - (b) Second Isomorphism Theorem
 - (c) Third Isomorphism Theorem
 - (d) Correspondence Theorem
2. Group actions
3. Cauchy's Theorem
4. p -Groups
5. Sylow Theorems
6. Simple groups
7. Basic ring theory
 - (a) Domains, integral domains, division rings, fields
 - (b) Subrings, direct products

You will be expected to state some common named theorems (e.g. First Isomorphism Theorem, Second Sylow Theorem, etc.). You won't be expected to state the Correspondence Theorem since its statement is a bit unwieldy, but you should be able to use it.

The exam will have 6 questions and 98 available marks (to earn full marks, you must get 95).